

Extrapolating Metal (Cu, Ni, Zn) Toxicity from Individuals to Populations Across *Daphnia* Species Using Mechanistic Models: The Roles of Uncertainty Propagation and Combined Physiological Modes of Action

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Abstract: Mechanistic effect modeling is a promising tool to improve the ecological realism of environmental risk assessment. An open question for the mechanistic modeling of metal toxicity is whether the same physiological mode of action (PMoA) could be assumed for closely related species. The implications of various modeling choices, such as the use of parameter point estimates and assumption of simplistic toxicodynamic models, are largely unexplored. We conducted life-table experiments with *Daphnia longispina*, *Daphnia magna*, and *Daphnia pulex* exposed to the single metals Cu, Ni, and Zn, and calibrated toxicokinetic–toxicodynamic (TKTD) models based on dynamic energy budget theory. We developed TKTD models with single and combined PMoAs to compare their goodness-of-fit and predicted population-level sensitivity. We identified the PMoA reproduction efficiency as most probable in all species for Ni and Zn, but not for Cu, and found that combined-PMoA models predicted higher population-level sensitivity than single-PMoA models, which was related to the predicted individual-level sensitivity, rather than to mechanistic differences between models. Using point estimates of parameters, instead of sampling from the probability distributions of parameters, could also lead to differences in the predicted population-level sensitivity. According to model predictions, achronic endpoints (cumulative reproduction, survival) are conservative for single-metal population effects across metals and species. We conclude that the assumption of an identical PMoA for different species of *Daphnia* could be justified for Ni and Zn, but not for Cu. Single-PMoA models are more appropriate than combined PMoA models from a model selection perspective, but propagation of the associated uncertainty should be considered. More accurate predictions of effects at low concentrations may nevertheless motivate the use of combined PMoA models. *Environ Toxicol Chem* 2024;43:338–358. © 2023 The Authors. *Environmental Toxicology and Chemistry* published by Wiley Periodicals LLC on behalf of SETAC.

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INTRODUCTION

Ecologically realistic risk assessment of metals in freshwater ecosystems is a complex challenge for multiple reasons,

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including the natural variability in responses of organisms to metal exposure (Bossuyt & Janssen, 2005; Myssen et al., 2005) and the need to make assessments of possible effects on field populations based on individual-level laboratory data. Simultaneously, scientists and regulators are increasingly acknowledging that mechanistic effect models are valuable tools to support environmental risk assessment (ERA) with such challenges.

One type of mechanistic effect model that may be used to tackle such challenges are toxicokinetic–toxicodynamic models based on dynamic energy budget theory (DEB-TKTD).

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studies by Ashauer & Jäger (2018) also indicated assimilation efficiency as a frequent PMoA for Cu. For Cd, in contrast, studies on two different nematode species have come to different conclusions about the PMoA (increase in growth costs in *Acrobeloides nanus* [Alvarez et al., 2006], decrease in assimilation efficiency in *Caenorhabditis elegans* [Swain et al., 2010]). A direct comparison between studies for metals can be hampered by different test conditions such as test media characteristics because it is currently unknown to what degree and how they will affect the PMoA. Different model selection procedures or details in the model formulation might also have implications for the concluded PMoAs. For most taxonomic groups and metals there is currently too little comparable information available that would allow conclusions to be drawn about the degree of species-specificity of PMoAs, and therefore the question of whether or not toxicity to closely related species could be modeled based on the same PMoAs deserves further investigation.

Consequences of PMoAs for modeling community-level effects. If closely related species vary in their respective PMoAs, this is of practical importance for modeling population- and community-level effects with DEB-TKTD models because PMoAs have implications for the resulting population dynamics (Vlaeminck et al., 2022) and therefore likely also for community-level effects. For example, Kooijman (2010) postulated that when populations are close to carrying capacity, most energy is spent on maintenance and therefore a stressor-induced increase in maintenance costs will more directly reduce carrying capacities than other PMoAs.

In general, a PMoA has to be inferred for each species (and each treatment, in the case of mixture exposure). However, if the same PMoA can be assumed for closely related species, this can greatly simplify the model development and reduce data requirements because a simple correction for the sensitivity of a species might be sufficient.

The most widespread model selection approaches assume a priori that sublethal toxicity is entirely caused by a single PMoA. Including only a single PMoA is desirable because it limits the number of TKTD parameters to be inferred from data. It may be necessary, however, to explore modeling approaches for calibrating DEB-TKTD models with combined PMoAs to accurately describe metal toxicity across the entire life cycle and range of sublethal concentrations. This is also motivated by the notion that metals act via mechanisms that could cause a wide array of effects on the physiological level (Grosell et al., 2002; Pane et al., 2003). In addition, a combination of multiple PMoAs explaining the data well might lead to different ratios of sensitivities across levels of organization, compared with the corresponding single-PMoA model. This aspect of DEB-TKTD model behavior has so far not been explicitly studied. In practice, model precision and accuracy have to be balanced with complexity, so the choice for a complex model should be underpinned by the relevance of the additional effects it can predict compared with the single-PMoA alternative, with respect to the intended model application. Identifying individual-level effects that are relevant for higher levels of

organization can be trivial in some cases (e.g., strong effects on survival), but may be non-trivial if effects are more subtle (e.g., effects on the timing of reproduction or hormetic effects). In the latter case, it is hard to make a reliable statement about whether effects are relevant without first calibrating the more complex model and assessing the differences in predicted effects in silico. It is therefore worthwhile to explore such studies and their implications for higher levels of organization.

Objective. The objective of the present study was to assess interspecific variability in PMoAs of the metals copper (Cu), nickel (Ni), and zinc (Zn) across *Daphnia* species (*D. longispina*, *D. magna*, and *D. pulex*), and assess the implications of two central modeling choices: the usage of a single point estimate to generate all predictions and the assumption of single-PMoA toxicodynamic models (each metal can only have one mode of action acting simultaneously).

The larger underlying problem definition is to extrapolate from single-metal individual-level toxicity to mixture toxicity on the community level in a variety of ecological scenarios. We conducted life-table experiments, exposing individuals of each species to the individual metals under identical conditions. We first calibrated single-PMoA TKTD models by performing a model selection between single-PMoA models, where in each model only one PMoA at a time is engaged. This allowed to assess interspecific variability more easily than when allowing for combined PMoAs. In addition, we calibrated the combined-PMoA models (metals can have multiple modes of action). We compared the two approaches with respect to goodness-of-fit and predicted population-level sensitivity relative to the most sensitive of two individual-level standard endpoints (21-day reproduction or survival). This highlights cases where individual-level endpoints may not be conservative for population level effects, and how sensitive general conclusions are to the tested modeling choices when using the models to extrapolate to untested conditions.

MATERIAL AND METHODS

Organism cultures

Isolated clones of the three species (*D. longispina* O.F. Müller 1785, *D. magna* Straus 1820, and *D. pulex* Leydig 1860) were obtained from the Laboratory of Aquatic Ecology, Evolution and Conservation, KU Leuven. The clones were originally sampled from Langeordevijver, Belgium (50.828581 N, 4.639960 E), and were cultured for >1 year in our laboratory before onset of the experiments.

Individuals were reared in COMBO medium (Kilham et al., 1998), modified to better reflect environmentally relevant conditions (15 mgCa L^{−1}, 5 mgMg L^{−1}) and avoid toxic levels of boron (1 mgL^{−1}). The medium was modified to better emulate environmentally relevant conditions. Culture media were titrated to pH=7.5 by adding HCl to a final concentration of 0.5 mM, aerated with oxygen stones, and allowed to equilibrate for at least 24 h before use. Cultures were daily fed an ad libitum amount of frozen (−80 °C) and defrosted suspension of *Raphidocelis subcapitata* cells. Culture media were renewed

Design and use of DEB-TKTD models

The DEB-TKTD models have previously been developed to extrapolate effects from the individual- to the population-level, and to better understand the links between individual- and population-level effects (Pereira, Vlaeminck, et al., 2019), and between single-metal and mixture toxicity on the population level (Hansul et al., 2021; Vlaeminck et al., 2021). Dynamic energy budget models have inherent properties that are useful for the extrapolation of stressor effects across levels of biological organization: DEB theory provides a single framework that can in principle be applied to all organisms, and DEB parameters can be estimated from individual-level data (Kooijman et al., 2008). For the extrapolation to the population level, DEB models can be integrated into population modeling approaches such as matrix population models (Billor et al., 2007; Klanjscek et al., 2006) and individual-based models (IBM, Martin et al., 2012). In particular, in combination with IBMs, a high level of flexibility can be achieved to simulate the dynamics of populations and communities.

As it has been demonstrated that DEB-TKTD models can be applied to explain and predict population-level effects, a logical next step is to apply this approach to the community level, requiring the calibration of such mechanistic effect models for different species under otherwise identical conditions. This step is needed for DEB-TKTD models and mechanistic effect models in general to advance them as tools to support ERA.

Estimation of TKTD parameters and physiological modes of action

Within the development of DEB-TKTD models, an important part is the identification of the physiological mode of action (PMoA). This is a DEB process through which the adverse effect of the metal in the model is caused. Mathematically, it determines how a chosen measure of the internal concentration of a chemical stressor is linked to state variables of the model. Frequent examples of this are an increase in the overhead costs for growth, an increase in maintenance costs, a decrease in assimilation efficiency, and an increase in overhead costs for reproduction. Other PMoAs are possible, however, such as a change in the somatic allocation fraction. The PMoA is usually identified indirectly in a model selection procedure based on how well alternative PMoAs explain individual level effects (Kooijman & Bedaux, 1996). This may be done by identifying the point estimates of parameters for each PMoA, identifying the PMoA based on model selection criteria, and generating all further predictions based on this estimate. Because parameter estimation is associated with uncertainty, so is the identification of PMoAs. Hence, another viable, but less frequently applied, approach is known as Bayesian model averaging (e.g., Genell et al., 2010; Thomas et al., 2007). In the context of DEB-TKTD models, the application of Bayesian model averaging translates to generating predictions based on all possible PMoAs and using the weighted average of predictions as the final prediction, where the weights are equal to the estimated probabilities of PMoAs.

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twice per week. Tests and experimental units were kept in a temperature-controlled room at 20 °C.

Life-table experiments

We conducted 21-day life-table experiments with each species, where individuals were exposed to a control at high food density (0.4 mg dry mass daphnid^{−1} day^{−1}), a control at reduced food density (compared with the tested high food density, 0.1 mg dry mass daphnid^{−1} day^{−1}) for estimation of DEB and starvation parameters, and five concentrations of Cu, Ni, and Zn each at high food for TKTD model selection and parameter estimation. The low food level was chosen so that a difference due to food availability in at least one measured variable (length, reproduction, survival) can be expected, but reproduction can still be observed. The purpose of the lower food level was to facilitate DEB parameter estimation (Kooijman et al., 2008), and in particular the surface-area specific filtration rate, which influences the relationship between ingestion rate and food concentration, and is therefore important for the application of DEB models in a population context. The lower food level might also contain valuable information on the minimum assimilation efficiency and adaptive plasticity parameters (see *Individual-based model*). The test medium was identical to culture medium, except for the addition of 4 mg natural dissolved organic carbon (DOC) L^{−1} instead of ethylenediaminetetraacetic acid. Natural DOC was used for the tests because natural DOC has different complexing behavior to artificial complexing agents. Natural DOC cannot be acquired in sufficient amounts to also add it to the culture media. Dissolved organic carbon treatment and source were identical to what is described by Hansul et al. (2021). Metal-spiked test media were allowed to equilibrate for at least 48 h.

Test animals were reared in 50 mL polyethylene jars filled with 40 mL of COMBO medium. Each experimental unit was initiated with a single 0 to 24-h-old third-brood neonate. Ten replicates per treatment were placed on wooden test plates in a randomized block design with two blocks of four and one block of two replicates per treatment, and the replicates of each treatment spread across three test plates. In the controls, carapace length was measured three times per week under a stereo microscope (Olympus SZX10) equipped with a camera (UC30). During preliminary experiments, we confirmed that the experimental handling imposed by length measurements did not significantly affect 21-day reproduction (p value > 0.05, one-sided t -test, data not shown). In all treatments, reproduction and mortality were recorded three times per week. On the same occasions, 100% of the test media were renewed. For *D. longispina* and *D. pulex*, carapace length was also measured in the metal exposures at day 21.

Temperature was recorded regularly in test vessels filled with deionized water.

For measurement of metal concentrations via inductively coupled plasma optical emission spectroscopy (ICP-OES), 10 mL of medium per treatment was taken as a pooled sample once per week (fresh medium at day 0 and old medium

As a model selection problem, the identification of PMoAs relies on replications of patterns by models. Within the context of individual-based models, the concept of pattern-oriented modeling has been established to aid in the development and evaluation of complex models (Grimm & Railsback, 2012). In the present study, we define the term “pattern” as any feature in the data that is considered to be biologically relevant and further relevant for the problem formulation under which the model is being developed. There does not appear to be a general recipe to apply pattern-oriented modeling, but it can be well integrated into a Bayesian inference procedure by considering the distinction between “strong” and “weak” patterns. A strong pattern will lead to correspondingly strong conclusions about parameter values or model structure (e.g., excluding a PMoA altogether). For a DEB-TKTD model, this might, for example, be the presence or absence of effects on growth. Weak patterns are harder to identify, tend to be of a more qualitative nature (e.g., a non-constant vs. constant effect strength over time), and lead to weaker conclusions about model structure and parameter values (e.g., narrowing or shifting the prior probability distribution). An example of a weak pattern might be the increase or decrease of a relative effect over time. Weak patterns are especially useful when used in combination because this tends to reduce the probability of a model predicting the correct pattern for the wrong reasons (Gallagher et al., 2021; Wiegand et al., 2004). Unifying pattern-oriented modeling with Bayesian inference naturally leads to the “empirical Bayes” approach (Casella, 1985). The purpose of empirical Bayes is to define informative or weakly informative prior distributions instead of uninformative priors, and the priors are estimated from the data itself instead of independent prior information. Empirical Bayes has been widely used in different fields (Carlin & Louis, 2000), but is not established for the calibration of DEB-TKTD models, or is only used implicitly.

Interspecific variability in PMoAs

Because it is known that sensitivity to metals, as expressed, for example, by an effect concentration 10% value, can vary by at least one order of magnitude (Bossuyt & Janssen, 2005; Myssen et al., 2005) between closely related species (e.g., within Cladocera), it is clear that TKTD parameters have to be estimated for each species to account for species-specific sensitivity. Less is known about the variability in PMoAs between closely related species (e.g., within the genus *Daphnia*), given identical test conditions and modeling approaches. Several studies have aimed to identify plausible PMoAs of metals for various freshwater organisms. Assimilation efficiency was put forward as the most plausible PMoA of Cu for both *Lymnaea stagnalis* (Vlaeminck et al., 2019) and *D. magna* (Vlaeminck et al., 2021). A DEB-TKTD model for Cu toxicity to *tigra* also included assimilation efficiency as a PMoA (Chen et al., 2012), but only for the juvenile stage, whereas effects on adult and larval stages were modeled by an increase in maintenance costs and an increase in growth costs, respectively. A summary of

at days 7, 14, and 21). Samples were 0.45 mL filtered, acidified with HNO₃ to a final concentration of 0.14 M, and analyzed using ICP-OES (ICAP 7000, Thermo Scientific).

For measurement of DOC, 10 mL pooled samples were taken from fresh media at day 0 and from old media at day 21, 0.45 mL filtered and analyzed with a total organic carbon analyzer (Shimadzu). Measured metal concentrations were converted to free ion activities with the Windermere Humic Aqueous model (WHAM) VII (Tipping et al., 2011) as described by Hansul et al. (2021), with the exception of an adjustment of the ΔLK2 value for Ni complexation from 1.57 to 2.35 based on Fettweis et al. (2021) and Van Laer et al. (2006). For all analyses, the average free-ion activity per treatment was used.

Individual-based model

We implemented an individual-based model based on DEB theory (DEB-IBM) using the Julia programming language. This implementation was used for all presented simulations. The overall goal of the model development was to extrapolate effects from the individuals to the community level (not part of the present study). For this purpose, effects of single metals on individual life history need to be modeled first. Some understanding of the predicted single-species population-level sensitivity will be necessary to understand the connection between predicted community dynamics and individual life history.

Individual life history was modeled according to the standard DEB model (Kooijman, 2010), with the addition of submodels for lethal and sublethal toxicity, and some adaptation with respect to assimilation and reserve investment, which are described below.

A full transparent and comprehensive ecological modeling (TRACE) documentation (Grimm et al., 2014), including detailed model description is provided as supporting information. Supporting Information, TRACE Tables S1 and S2, list all the implemented individual and global attributes.

Feeding and assimilation

Individuals feed from a resource p that is present at an external concentration K_p (J L^{−3}). The ingestion rate J_p (J day^{−1}) depends on the functional response $f(=)$, the maximum surface-area specific ingestion rate I_{max} (J cm² day^{−1}), and the individual's squared structural length L^2 (cm²).

$$J_p = f I_{\text{max}} L^2 \quad (1)$$

where f represents a type II-functional response in which the half-saturation constant K_x (J L^{−3}) is the ratio between I_{max} and the maximum surface-area specific filtration rate (f_{max}) (L cm^{−2} day^{−1}; Martin et al., 2012).

$$f = \frac{X_p}{K_x + X_p} \quad (2)$$

Equations (1) and (2) are identical to previous DEB-IBM implementations (e.g., Martin et al., 2012; Vlaeminck et al., 2021).

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The assimilation rate β_A (J day⁻¹) was modeled as the product of ingestion rate and assimilation efficiency $K_{E,X}$ (–):

$$\beta_A = J_{E,X} K_{E,X} \quad (2)$$

For *Daphnia*, $K_{E,X}$ tends to be higher at lower food concentrations (Urabe & Watanabe, 1991). We implemented this effect following Rinke and Vijverberg (2005):

$$K_{E,X} = K_{E,X,max} - (K_{E,X,max} - K_{E,X,min})f \quad (4)$$

The maximum assimilation efficiency $K_{E,X,max}$ was set to 0.9 (Rinke & Vijverberg, 2005) and the minimum assimilation efficiency $K_{E,X,min}$ was estimated from control data with an upper limit at 0.9. Because assimilation efficiency can vary by a factor of 5 as a function of the food level (Urabe & Watanabe, 1991), we suspect that the implementation of a food-dependent assimilation efficiency is highly relevant for the accurate description of the response of *Daphnia* individuals to food limitation, and therefore also relevant for the prediction of population dynamics.

The inclusion of an adaptive plasticity submodel causing an increase in the somatic investment ratio k at decreased energy density (Gergs et al., 2014) was considered. The change in k was modeled as a log-logistic dependency on the scaled reserve density e :

$$k = k_0 + \frac{1 - k_0}{1 + \left(\frac{e}{e_{0.5}}\right)^{\beta_{k,e}}} \quad (5)$$

where k_0 is the somatic allocation fraction at $e = 1$ (situation for a well-fed animal), e is the scaled reserve density, where reserve density is the amount of reserve relative to the amount of structure, and the scaling is done by the maximum reserve density (see section Supporting Information, TRACE Model description, for more details). $e_{0.5}$ is the value for e at which k is equal to the mean between k_0 and 1, and $\beta_{k,e}$ the available life table data contain information on plausible values of $e_{0.5}$ and $\beta_{k,e}$ if growth is less affected and simultaneously reproduction is more affected by decreases in food density than predicted by standard DEB. Otherwise, $e_{0.5}$ is likely left-censored. The implementation of an adaptive plasticity component is justified by the finding that adaptive plasticity greatly improves the predictive performance of DEB-BMs in *Daphnia* populations (Gergs et al., 2014). Adaptive plasticity is mostly relevant for simulation of population and community dynamics, but can have an effect on TKTD parameter probabilities in combination with PMoA assimilation efficiency.

Starvation mortality

We modeled increased mortality at decreased energy availability as a stochastic mortality probability depending on e , similar to previous DEB-BM implementations (Martin et al., 2013; Vlaeminck et al., 2021), but deviated from the assumption of a linear relationship between e and the starvation-induced mortality probability. Instead, we related e to the

reserve-dependent survival probability β_S using a shape parameter β_S (–):

$$\beta_S = \begin{cases} \exp(\ln(e)\beta_S) & \text{if } e < e_{0.5,\beta_S} \\ 1 & \text{otherwise} \end{cases} \quad (6)$$

This means that higher values for β_S and $e_{0.5,\beta_S}$ imply higher susceptibility to death from reduced energy reserves. Any species will be guaranteed to have a survival probability of 0 when energy reserves are 0, and a starvation-related survival probability of 1 when $e = 1$ (given $0 < e_{0.5,\beta_S} < 1$). In contrast to a strictly linear relationship, Equation (6) helps with matching particularly steep-sloped decreases in survival as a function of reserve density without implying mortality at high e .

TKTD submodel

The TKTD submodel included a simple toxicokinetic component, where external concentration of metal z , C_z (nM free-ion activity), is translated to a scaled damage $D_{z,i}$ (nM free-ion activity), depending on a single toxicokinetic parameter, the dominant rate constant $k_{D,z,i}$ (day⁻¹; Jager et al., 2011):

$$\frac{dD_{z,i}}{dt} = k_{D,z,i}(C_z - D_{z,i}) \quad (7)$$

$D_{z,i}$ has the same unit as an external concentration because it is scaled by the bioconcentration factor (Jager et al., 2011). It is interpreted as being proportional to the bioavailable concentration at the target site and is therefore specific to PMoA j .

We initially considered four PMoAs, namely growth costs (G), increase in the amount of energy required for somatic growth, also interpretable as decrease in growth efficiency), maintenance costs (M, increase in somatic maintenance rate), assimilation efficiency (A, decrease in assimilation efficiency), and reproduction efficiency (R, decrease in reproduction efficiency). For cases where the four initially considered PMoAs (individually or combinations thereof) could not explain observed effects, we also included a novel PMoA, called “embryonic investment.” This PMoA causes a decrease in initial energy of an embryo, E_e (the default value for E_e is calculated internally from DEB parameters, see TRACE for details). It thus provides a mechanism for increased brood sizes without affecting growth, and without implying an increase in individual fitness under metal stress. The same effect can usually not be explained with increases in reproduction efficiency because reproduction efficiency is already high in the controls (default value of 0.95; Koijman & Gergs, 2019).

For each PMoA j , $D_{z,i}$ was related to a relative response $Y_{z,j}$ (–) by a three-parameter log-logistic equation with parameters maximum stress $s_{max,z,j}$ (–), median effective damage $ED_{50,z,j}$ (nM), and Hill slope $\beta_{z,j}$ (–). For effects via PMoAs G and M, the physiological dose-response curve has a lower limit of 1 and increases with increasing $D_{z,i}$:

$$Y_{z,j} = 1 + \frac{s_{j,max}}{1 + \left(\frac{D_{z,i}}{ED_{50,z,j}}\right)^{\beta_{z,j}}} \quad (8)$$

DEB and starvation parameter estimation

The starvation-related hazard rate and the DEB parameters were fitted simultaneously to carapace length, cumulative reproduction, and survival over time in the controls at two food densities. For this purpose, observed and predicted values were scaled by the maximum of observed values of each observed variable. As a distance function, we used the mean squared deviation (MSD), calculated for each food level and variable, and summed up to yield the total deviation ΣMSD . Note that ΣMSD is comparable across models for the same species/metal combination, but not across species and metals. Parameter values for the three *daphnia* species are reported in the Add my Pet database (Koijman & Gergs, 2019), but predictions based on these were very different from our observations (Supporting Information, Table S4 and Figure S1), likely due to the implemented adaptations in the DEB model. We therefore initially conducted a manual adjustment of DEB parameters, focusing on most influential parameters (Supporting Information, Table S5) and used the adjusted values as prior means for ABC. Priors were set to normal distributions with a coefficient of variation of 0.5. Using broader priors was not found to be useful because the majority of parameter samples were invalid, which means they returned a prediction for which the deviation from the data could not be evaluated, for example because simulated individuals would not even be able to transition from embryonic to juvenile stage. To conduct the ABC, we evaluated 100,000 posterior samples and accepted the best 1% of valid parameter values. The posterior evaluation of the fitted DEB, starvation, and adaptive plasticity submodels was conducted by sampling 100 values from the posterior distribution and generating the prediction for each sample. The reported predictions are the mean and given percentiles of all predicted values. The prediction interval encompasses epistemic uncertainty (uncertainty in parameter estimation) and aleatory variability (stochastic mortality, individual variability). The uncertainty in these parameters was also propagated during estimation of TKTD parameters.

TKTD parameter estimation

For each species–metal combination, we compared two approaches of model calibrations. First, we fitted the single-PMoA models (one PMoA at a time) and conducted a model selection. Second, we fitted models including the combination of all PMoAs that were not excluded based on the prior statistical analysis (combined-PMoA models).

Priors for the physiological-level sensitivity (median effective damage $ED_{50,z}$) were set using an empirical Bayes approach (Casella, 1985). All priors were defined as truncated normal ($\hat{\mu}_{0,z}, \hat{\sigma}_z$) or log-normal ($ED_{50,z}$) distributions.

For each species/metal distribution, our approach for the TKTD parameter estimation can be divided into four steps: excluding PMoAs a priori where appropriate, defining prior distributions for the physiological-level sensitivities, defining prior distributions for the remaining parameters, and running the ABC.

The first step was to identify PMoAs that could be excluded from the estimation, based on a simple statistical analysis of the data.

For this purpose, we used two strong patterns (presence/absence of effects on final length or final cumulative reproduction). We tested for significant effects on final length, final cumulative reproduction (up to the last time point with survival > 0), as well as the control-normalized average brood size (Monte-Carlo permutation test for difference in mean relative response, 10^3 permutations per test). When calculating the final cumulative reproduction, the cumulative reproduction of adult test animals that died during the test was included by comparing their cumulative reproduction with the average cumulative reproduction in the control at the time point of death. This differs from standard reproductive endpoints (Organisation of Economic Co-operation and Development [OECD], 2012) because the purpose was to differentiate between mechanistic effects on reproduction and effects on survival. In cases where final length was measured in the metal treatments, but no effect observed, PMoA maintenance costs and assimilation efficiency were excluded (Figure 1). In cases where no effect on final cumulative reproduction was observed, PMoA maintenance costs, assimilation efficiency, and reproduction efficiency were excluded (Figure 1). We did not exclude PMoA growth costs when observing no effects on 21-day carapace length because the growth costs do not affect equilibrium conditions and we aimed to be cautious to exclude PMoAs altogether from the ABC-based inference. While it is likely that some effect on length and reproduction is observed when there are effects on growth costs, it is possible that they are very small.

The second step in the TKTD estimation was to define plausible prior means for the $ED_{50,z}$ parameters.

For this purpose, we used a single weak pattern, namely effects on average brood size without effects on cumulative reproduction. In other words, a decrease in the relative effect on cumulative reproduction over time (cf. growth costs in Figure 1). If average brood size is affected, but final cumulative reproduction is not, this effect is most likely driven by PMoA growth costs (reproduction is initially affected but effects become very small). In this case, the prior median for $ED_{50,z}$ was set to the geometric mean of test concentrations in which these conditions were met. The $ED_{50,z}$ values for the remaining PMoAs were set to the geometric mean of the range of test concentrations within which any effects on reproduction (average control-normalized brood size or final cumulative reproduction) were observed. The standard deviation of prior distributions for $ED_{50,z}$ parameters was by default set to 1 on a naturally logarithmic scale.

The presence of an adverse effect was defined as being both statistically significant and falling below a relative response threshold of 0.95. Relative responses below 0.5 were also considered effects, regardless of the p value.

After excluding PMoAs and setting prior distributions for the $ED_{50,z}$ parameters, the prior distributions for the remaining parameters ($k_{D,z,i}$, $\beta_{z,j}$, $s_{max,z,j}$) had to be defined.

To simplify the calibration, we used a narrow prior for the toxicokinetic parameter by default (normal distribution with

For effects via PMoAs A and R, the physiological dose-response curve has an upper limit of 1 and decreases with increasing $D_{z,i}$:

$$Y_{z,j} = 1 - \frac{s_{j,max}}{1 + \left(\frac{D_{z,i}}{ED_{50,z,j}}\right)^{\beta_{z,j}}} \quad (9)$$

where $Y_{z,j}$ is the control-normalized value of the state variables affected by metal z through PMoA j , that is, the relative response on the physiological level represent a deviation from the classic DEBtox approach (Jager et al., 2006) by assuming a log-logistic dose response on the physiological level instead, and by use of a maximum stress parameter. The more common function in general unified threshold model of survival (GUTS) and DEB-TKD models is a linear dependency between stress on damage above a certain threshold, which has been justified with the notion that each additional molecule of the chemical stressor can be assumed to have the same effect on hazard rate (Jager et al., 2011) and, by extrapolation, physiological stress.

There is, however, no clear connection between this reasoning and biochemical aspects of toxicodynamics, that is, enzyme and receptor-ligand kinetics. For example, if a metal (or any other chemical stressor) acts as enzyme inhibitor, it is plausible to expect that the effect will saturate as a larger fraction of enzymes are bound by metal ions. Because the log-logistic equation is the common default choice in ecotoxicology, we argue that it is at least equally plausible to assume a log-logistic dose-response function by default for GUTS and DEB-TKD models. Figure 1 illustrates the effects on growth and reproduction following from each of the four initially considered PMoAs for a hypothetical example.

In addition to sublethal effects, effects on survival which could not be explained through effects on DEB processes were modeled with a hazard rate h_z (day⁻¹) which was related to $D_{z,i}$ by a three-parameter log-logistic equation with lower limit 0 and upper limit h_{max} (day⁻¹):

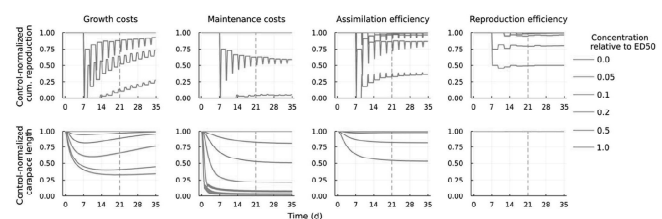


FIGURE 1 Effects of physiological modes of action on reproduction (expressed as control-normalized cumulative reproduction over time) and growth (control-normalized carapace length over time) for a hypothetical example and in the absence of (direct or indirect) effects on survival. The dashed vertical line indicates the duration of the conducted life-table experiments. The appearance of the relative response for cumulative reproduction arises from discrete reproduction (broods are released in discrete time intervals; see Supporting Information, TRACE Model description, for details). ED_{50} = median effect dose.

mean 1 and standard deviation 0.1) because this is likely to be one of the less important parameters (Billiot et al., 2008). We further fixed the $s_{max,z,j}$ values to 100 for growth costs and maintenance costs, and 1 for assimilation efficiency and reproduction efficiency. This means that any PMoA will eventually cause high stress if the damage is high enough, and that the relative contribution of PMoAs in combined-PMoA models will be chiefly controlled by the $ED_{50,z}$ and $\beta_{z,j}$ parameters. The priors for $ED_{50,z}$ parameters were set broad enough so that it was possible for any PMoA to be cancelled out of the model during calibration of combined-PMoA models because of very low estimated sensitivity. For growth costs, a $s_{max,z,j}$ value of 100 is interpretable as a maximum 99% reduction in growth efficiency. The empirical Bayes algorithm that was used to derive the priors for all species and metals is available as Julia script on Figshare.

After setting all the prior distributions, we conducted the ABC by calculating 250,000 samples per PMoA for single-PMoA models (i.e., 10^6 in total if all four PMoAs were included) and 10^6 for combined-PMoA models, and accepted the best 0.1% of parameter values for each species/metal combination. In the case of single-PMoA models, the set of accepted samples therefore includes samples from different PMoAs in different frequencies. The combined-PMoA models allowed for simultaneous action of all PMoAs that were not excluded a priori. Note that this does not imply that all PMoAs combined do cause the effect in the calibrated combined-PMoA models

because the estimated sensitivity associated with some PMoAs could be very low.

Posterior evaluation of calibrated TKTD models

During posterior evaluation of the calibrated TKTD models, we compared the two different types of TKTD models (single-PMoA vs. combined-PMoA), and two approaches to generate predictions from calibrated models: identifying a point estimate (best fit) or sampling from the probability distributions of PMoAs and corresponding parameters to derive an aggregate prediction with account for uncertainty. The best fit is the parameter combination associated with the lowest distance to the data (as returned by ABC). This resulted in three different predictions, best fit, single-PMoA model average, and combined PMoAs (Figure 2), where the aim was to compare the single-PMoA model average with the best fit prediction to assess the implications of propagating uncertainty and to compare the single-PMoA model average with combined-PMoA models to assess the implications of accounting for the possibility of multiple PMoAs explaining observed effects.

For single-PMoA models, we calculated the relative probability of PMoAs as the number of accepted parameter samples per model relative to the total number of samples (Mertens et al., 2018). We evaluated the goodness-of-fit of calibrated models quantitatively and visually. For quantitative evaluation,

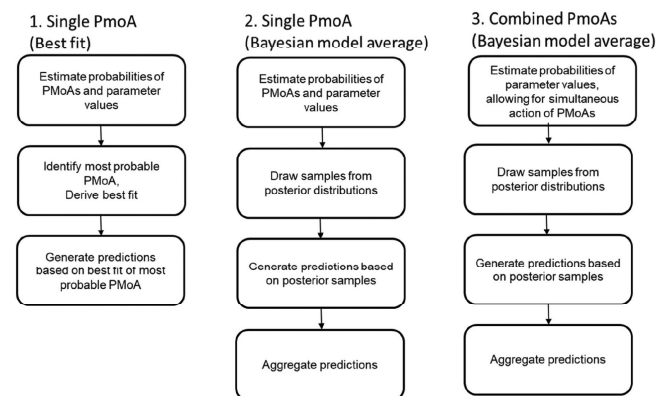


FIGURE 2 Three ways to calibrate toxicokinetic-toxicodynamic models based on dynamic energy budget theory and generate predictions from calibrated models were used in the present study. PMoA = physiological mode of action.

we calculated the mean absolute error (MAE), normalized Nash-Sutcliffe efficiency (nNSE), and a posterior predictive check (PPC) for each measured variable. The MAE indicates precision and accuracy and was preferred over mean relative error due to the large proportion of zeros in reproduction and survival data.

$$\text{MAE} = \frac{\sum_{i=1}^N |O_{it} - P_{it}|}{N} \quad (11)$$

where O_{it} is the i th observed value, P_{it} is the i th predicted value of an observed variable and N is the number of observations. The Nash-Sutcliffe efficiency (Nash & Sutcliffe, 1970) compares the goodness of fit with that of the mean of all observed values and ranges from $-\infty$ to 1, where an Nash-Sutcliffe efficiency of 1 corresponds to a perfect prediction and an Nash-Sutcliffe efficiency of 0 corresponds to a prediction that is equally close to the data as the mean of all observations.

$$\text{nNSE} = 1 - \frac{\sum_{i=1}^N (O_{it} - P_{it})^2}{\sum_{i=1}^N (O_{it} - \bar{O}_{it})^2} \quad (12)$$

nNSE re-scales NSE to a value between 0 and 1:

$$\text{nNSE} = \frac{1}{2 - \text{NSE}} \quad (13)$$

The PPC applies specifically to probabilistic predictions and is the fraction of observations falling within the 5th to 95th percentile range of observed values. Because we calculated these criteria for each variable separately and for the purpose of direct interpretation, we did so based on the raw data points instead of scaled and $\log_{10}$ -transformed values as done for the parameter inference itself.

We conducted a visual predictive check by plotting predicted and observed cumulative reproduction, control-normalized brood sizes, carapace length, and survival over time. In addition, we visually evaluated observed and predicted concentration-response curves for 21-day cumulative reproduction of surviving female parent animals, 21-day carapace length, and 21-day survival.

Extrapolation to the population level

To assess population-level sensitivity, we used the calibrated DEB-TKTD models to simulate 8-week population dynamics at the same exposure concentrations that were tested in the life-table experiments (average free-ion activities calculated from measured dissolved concentrations). As for the posterior evaluation, we conducted the extrapolation in three different ways (best fit, single-PMoA model average, combined PMoA) to evaluate the potential implications of these modeling choices. The simulations of population dynamics mimicked a laboratory microcosm experiment with daily food addition (0.4 mg dry mass $\text{L}^{-1} \text{day}^{-1}$) in 500 mL of medium, a constant loss rate of food due to algal cell sinking (0.5 day^{-1}) and bi-weekly 25% renewals of test medium (proportional

removal of residual food in the test medium). The capacity of DEB-IBM models to predict population dynamics and effects of metals in similar scenarios has been demonstrated previously for Ni (Pereira, Blust, et al., 2019), Cu, and Zn (Vlaemink et al., 2021) in *D. magna*. As population-level endpoints, we used a carrying capacity (daphnids L^{-1}), approximated as population density averaged from day 49 to 56) and the initial growth rate. The initial growth rate μ was calculated based on the peak density N_{peak} , initial density N_0 , and the time to reach peak density t_{peak} (Hansul et al., 2021):

$$\mu = \begin{cases} \frac{\ln(N_{\text{peak}}) - \ln(N_0)}{t_{\text{peak}}} & \text{if } N_{\text{peak}} \geq N_0 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

μ was set to 0 for cases where populations go extinct because in the simulated scenario extinction will, if at all, occur within a short amount of time, so there is little added value in accounting for different times to reach extinction.

We further calculated the difference between population-level and individual-level sensitivity based on the predictions of the corresponding model at each exposure concentration. Because this calculation is only based on predicted values, it allows the identification of differences between predicted population-level sensitivities that are caused by mechanistic differences between the models, rather than different predicted individual-level sensitivity. As a reference for individual-level sensitivity, we used the minimum relative response (21-day cumulative reproduction of surviving parent animals or survival) at the corresponding exposure concentration. This further allowed the identification of cases where individual-level endpoints may not be conservative for population-level effects.

RESULTS

Measured test media characteristics and control performance

Test media characteristics including water temperature are reported in Supporting Information, Table S1 and Figure S1. Details on detection limits and quality control of ICP-OES measurements are given in Supporting Information, Tables S2 and S3. The pH (8.0 instead of 7.5) was slightly higher than nominal, whereas average DOC (3.2 mg L^{-1} instead of 4.0 mg L^{-1}) was lower than nominal (Supporting Information, Table S1). Average temperature was 1 °C higher than nominal (21 °C instead of 20 °C). Mean measured dissolved concentrations did not deviate by more than a factor of 1.8 from nominal concentrations. All WHAM input and output files are available on Figshare.

In the controls, the average final cumulative reproductive output of surviving parent animals was 58 (SD = 15, $n = 7$), 36 (SD = 14, $n = 10$), and 72 (SD = 16, $n = 8$) for *D. longispina*, *D. magna*, and *D. pulex*, respectively, with a final average carapace length of 0.18 cm (SD = 0.007), 0.3 cm (SD = 0.013), and 0.25 cm (SD = 0.012). The 21-day mortality at high food was 0% in *D. magna* and *D. pulex*, and 30% in *D. longispina*.

DEB and starvation parameter estimation

By estimating DEB parameters, individual life history at two food levels could be well approximated for all three species (Supporting Information, Figure S2). Compared with values reported in the Add my Pet database, multiple parameters (Supporting Information, Table S6) needed to be corrected for all species to achieve a good fit. For the shape parameter (conversion from structural length to carapace length), we decided to use the same value for all species. This may be crucial for application of the parameterized DEB models in a community context because differently shaped parameters for different species can result in different relative body sizes when expressed in structural length instead of carapace length (i.e., the largest species in terms of carapace length may not be the largest species in terms of structural length). We adopted a value of 0.211 for the shape parameter. This value was taken from Vlaemink et al. (2021) and used because it was clear that carapace length in that study was measured in the same way as in the present study.

A common pattern in the estimation of DEB parameters was a narrower (compared with the prior) marginal posterior distribution for the parameters maximum surface area specific ingestion rate (I_{max}), minimum assimilation efficiency (K_{min}), somatic investment ratio (k_1), and volume-specific maintenance costs (p_{m} , p_{v}) (Supporting Information, Figures S3–S5). For the remaining DEB parameters, the data only provided weak information. Species noticeably differed in their response to reduced food, as shown by *D. longispina* suffering from a greater relative increase in mortality compared with the background mortality at high food.

Prior analysis of toxicity data and pre-selection of PMoAs

The results of the prior analysis are summarized in Supporting Information, Tables S7 and S8.

In *D. longispina*, Cu did not adversely affect reproduction (neither the final cumulative reproduction, nor the control-normalized average brood size; Figure 3) or 21-day carapace length (Figure 4). Copper did, however, affect survival of *D. longispina* up to 100% at the two highest tested concentrations (15 and 70 nM, Figure 6). The TKTD model for Cu toxicity to *D. longispina* was therefore reduced to direct lethal effects. Nickel did not affect 21-day carapace length of *D. longispina* at any tested concentration (Figure 4) so the PMoAs maintenance costs and assimilation efficiency were excluded. Zinc affected 21-day carapace length of *D. longispina* at 450 nM (Supporting Information, Table S7 and Figure S5), affected mean control-normalized mean brood size at 91 nM and 160 nM, and affected final cumulative reproduction at 91 nM and above (Supporting Information, Table S8 and Figure S4). For Zn toxicity to *D. longispina*, no PMoAs were therefore excluded a priori.

In *D. magna*, no PMoAs were excluded a priori from any of the TKTD models. Because the analysis relied only on reproduction and survival data, the definition of prior distributions

and parameter inference relied on weak patterns. Other than in *D. longispina* and *D. pulex*, all three metals affected final cumulative reproduction of *D. magna* (Figure 3 and Supporting Information, Table S8).

In *D. pulex*, we observed no effects of any metal on final cumulative reproduction of surviving parent animals (Figure 6 and Supporting Information, Table S8), but did observe effects on mean control-normalized mean brood size. Effects on brood size followed a pattern characterized by an up to 2.5-fold increase in brood sizes, which tended to occur in later adult instars (Supporting Information, Figure S6) and is not predictable by adverse effects with the PMoAs shown in Figure 1. In Cu and Ni exposures, adverse effects on brood sizes occurred in earlier adult instars at the same concentrations where increases in brood sizes were observed. To fit this pattern, we included the PMoA embryonic investment for *D. pulex*. For the PMoA embryonic investment, we allowed for slower toxicokinetics (prior mean $k_{\text{DEB}} = 0.1$) because the corresponding pattern (increase in brood size) established gradually over time.

TKTD model calibration

PMoA probabilities. For Ni and Zn, reproduction efficiency was the most probable PMoA in all species, albeit with varying degrees of evidence (Figure 5). For Ni toxicity in *D. longispina* and *D. magna*, growth costs were only slightly less probable than reproduction efficiency (50 percentage points difference).

For Cu, the identity of the most likely PMoAs varied between species. Cu toxicity to *D. longispina* was modeled based on direct lethal effects only. In *D. magna*, PMoAs reproduction efficiency and assimilation efficiency were most likely (36% and 34% probability, respectively), while growth costs could also not be ruled out entirely as plausible PMoA (25% probability). For Cu toxicity to *D. pulex*, embryonic investment was identified as most probable PMoA (65% probability). Notably, PMoA maintenance costs were never among the most likely PMoAs.

Goodness-of-fit and parameter estimates

Based on global quantitative evaluation (nNSE, MAE, PPC), there was no general trend as to which TKTD model (single-PMoA vs. combined-PMoA) and prediction type (single-PMoA model average vs. best fit) predicted the data better (Supporting Information, Figures S8–S10). For *D. pulex*, fixed values for $S_{\text{max},Z}$ did not lead to a good fit, especially for Cu and Zn. The calibrations for *D. pulex* were therefore repeated with variable values for $S_{\text{max},Z}$. Furthermore, we opted to use narrower priors for *D. pulex* (Supporting Information, Figures S23–S24, S29–S30, and S33–S36) to be able to fit the specific temporal pattern of effects on brood sizes (Supporting Information, Figures S13, S16, and S19). Across species, metals, and response variables, the average ratio of goodness-of-fit between single-PMoA and combined PMoA models was 1.0 based on both nNSE (SD = 0.061, $n = 29$) and PPC (SD = 0.055, $n = 29$). The biggest differences were found in *D. pulex*, but, on average, model types still fitted the data equally well.

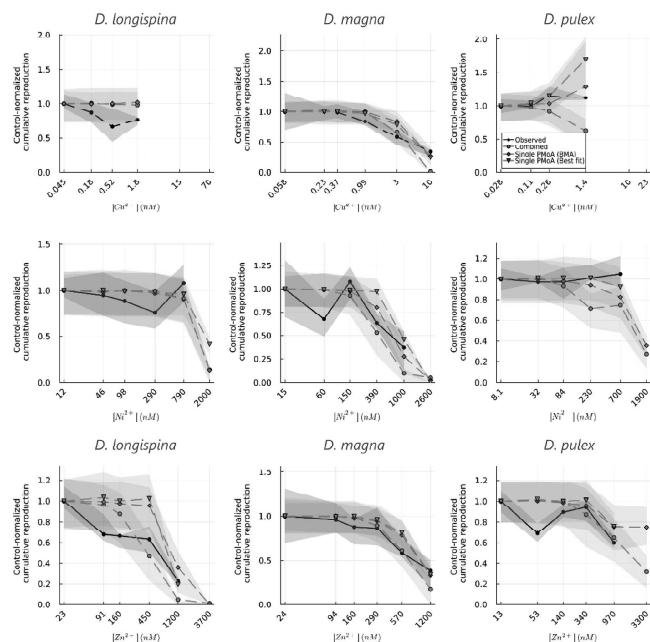


FIGURE 3 Model fits to cumulative reproduction (reproductive output per surviving female parent animal at the end of the test). Error bands indicate interquartile ranges. Prediction errors reflect model stochasticity, parameter uncertainty, and uncertainty in selection of physiological modes of action (PMoA). Cu toxicity to *Daphnia longispina* was modeled solely based on direct lethal effects, so only the comparison between Bayesian model average (BMA) and best fit is made. Concentrations without values plotted caused 100% mortality in observations and all predictions.

Visual evaluation showed that combined-PMoA models overall tended to overpredict sublethal effects at high effect concentrations, whereas single-PMoA models tended to underpredict effects at low effect concentrations. This led to a better prediction of low-effect levels to cumulative reproduction (reproductive output per surviving parent animal at the end of the test) by the combined-PMoA models for Cu toxicity to *D. magna* and Zn toxicity to all species (Figure 3 and

Supporting Information, Figures S11–S19), but (as confirmed by quantitative evaluation) not an overall better fit. A clear example for this trend is given by Zn toxicity to *D. magna*. At 570 nM Zn^{2+} , the combined-PMoA accurately and precisely predicted the effect on cumulative reproduction, whereas the single-PMoA models clearly underpredicted the relative effect. At 1200 nM Zn^{2+} , the combined-PMoA model overpredicted the relative effect, whereas the single-PMoA model predicted

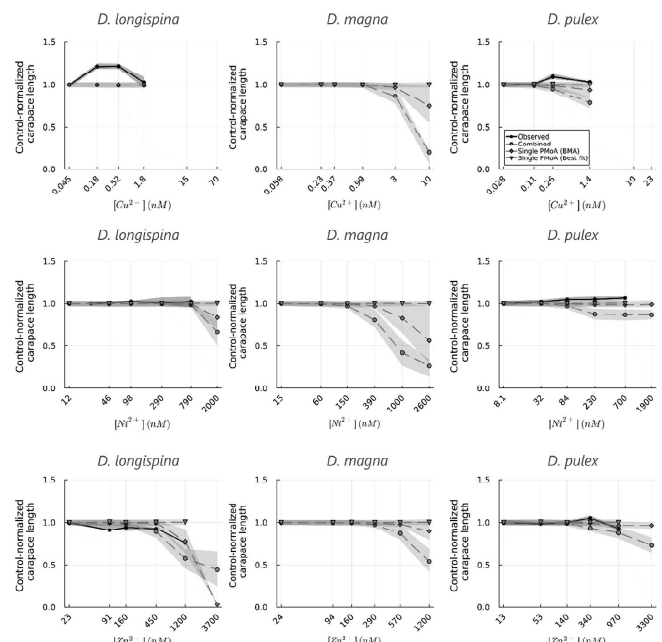


FIGURE 4 Model fits to relative responses of 21-day carapace length (*Daphnia longispina* and *Daphnia pulex*) and predicted relative responses of 21-day carapace length (*Daphnia magna*). Error bands indicate interquartile ranges. Prediction errors reflect model stochasticity, parameter uncertainty, and uncertainty in selection of physiological modes of action (PMoA). Concentrations without values plotted caused 100% mortality in observations and all predictions. BMA = Bayesian model average.

the effect well. Similar trends can, for example, be seen in the fits of final cumulative reproduction for Zn toxicity to *D. longispina* and Ni toxicity to *D. magna*.

The visual predictive check revealed for Ni and Zn toxicity to *D. pulex* that only combined-PMoA models accurately predicted the temporal pattern of control-normalized brood sizes (increase in the relative response over time), whereas the single-PMoA models predicted a constant effect over time (Supporting Information, Figures S16 and S19).

Effects on growth and survival were differently predicted by the different model and prediction types, but with no general trend with respect to accuracy and bias (Figures 4 and 6). Model predictions for carapace length diverged more clearly in cases where carapace length was not measured (*D. magna*), with combined-PMoA models generally predicting stronger effects on growth (Figure 4).

Similarly to combined-PMoA models, the global goodness-of-fit based on the best fit (estimates shown in

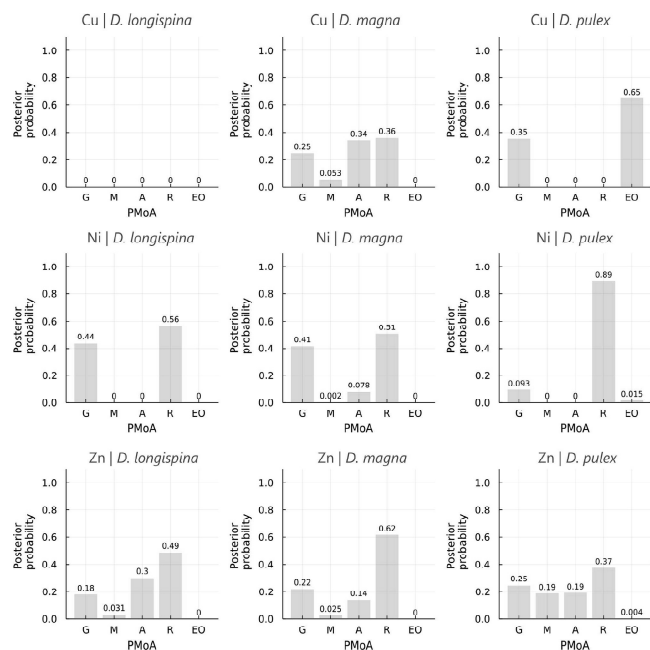


FIGURE 5: Relative probabilities of physiological modes of action (PMoAs) for single-PMoA models. PMoAs with probability $\neq 0$ were excluded based on the prior statistical analysis. Cu toxicity to *Daphnia longispina* was modeled solely based on direct lethal effects, hence all PMoAs were excluded. PMoAs: G = increase in growth costs; M = increase in maintenance costs; A = decrease in assimilation efficiency; R = decrease in reproduction efficiency; EO = decrease in the amount of energy invested per offspring individual.

Supporting Information, Table S9) was overall close to that of the single-PMoA model average (Supporting Information, Figures S7–S10). Visual evaluation showed qualitatively different predictions but no general trend across species and metals.

All prior and posterior distributions are shown in Supporting Information, Figures S20–S36. These show that in many cases the marginal posterior distributions for $k_{0.2,2}$ parameters were very close to the prior distributions. That means the effect

data contained little information on toxicokinetics within the range of considered values, and the estimates were largely determined by the prior distributions.

Extrapolation to the population level

Population dynamics were predicted to follow a delayed logistic growth pattern, as previously observed in comparable scenarios (Preuss et al., 2009; Vlaeminck et al., 2021;

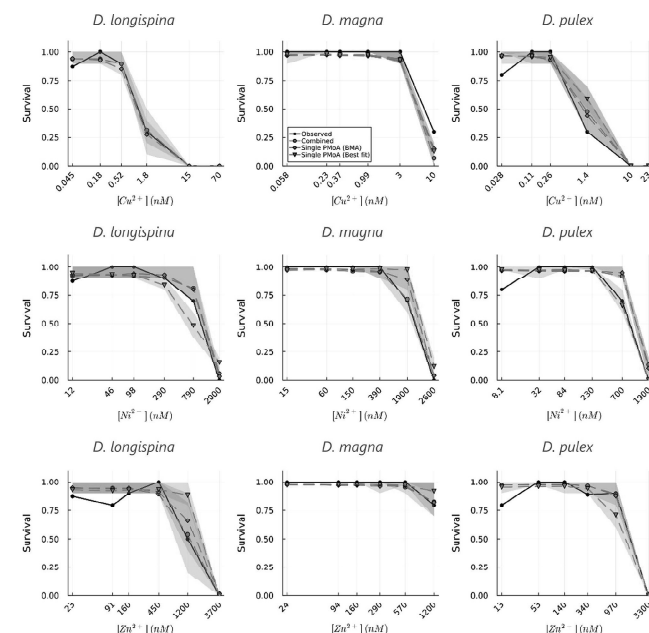


FIGURE 6: Model fits to 21-day survival. Error bands indicate interquartile ranges. Prediction errors reflect model stochasticity, parameter uncertainty, and uncertainty in selection of physiological modes of action (PMoAs). BMA = Bayesian model average.

Supporting Information, Figures S37–S39). In the metal treatments, the predicted responses of population growth rate showed that combined PMoA models tended to predict overall higher population-level sensitivity (Figure 7 and see Supporting Information, Figure S40, for corresponding effects on growth rates). Across species and metals, the average ratio between predicted relative responses of growth rates by combined-PMoA models compared with the single-PMoA model average was 0.87 (SD = 0.29, $n = 47$), but the deviation was much higher in individual cases (e.g., Ni toxicity to *D. magna*, Zn toxicity to *D. longispina*). In *D. longispina* and *D. magna*, these findings were independent of the choice of endpoint, whereas in

D. pulex, the relationship was inverted for carrying capacity (higher sensitivity predicted by single-PMoA models, Figure 7), which coincides with inclusion of the PMoA embryonic investment. Because the combined-PMoA models also tended to predict higher sensitivity on the individual level (average brood sizes and cumulative reproduction), the comparison of relative population-level responses alone does not allow the conclusion that this is the result of different population dynamics emerging from combined PMoAs. The prediction based on the best fit was often close to that of the single-PMoA model average, but differed clearly in specific cases. Extreme examples thereof are given by predicted effects of Ni on *D. pulex* carrying capacity

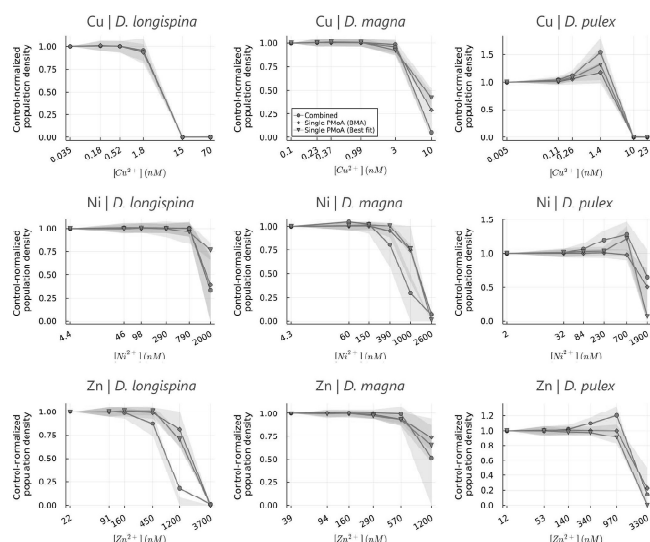


FIGURE 7: Predicted effects of metals on carrying capacity, based on single-physiological modes of action (PMoAs) and combined PMoA models. BMA = Bayesian model average.

(93% effect vs. 49% effect at 1900 nM based on best fit and single-PMoA model average, respectively; Figure 7) and Ni toxicity to *D. longispina* growth rate and carrying capacity (35% effect vs. 73% effect at 2000 nM based on best fit and single-PMoA model average, respectively; Figure 7 and Supporting Information, TRACE Figure S40).

Compared with the predicted individual-level sensitivity, all models predicted populations to be equally or less sensitive, independent of the population-level endpoint (Figure 8 and Supporting Information, TRACE Figure S41). The differences between predictions were overall smaller than when comparing the relative responses directly, particularly for the comparison between single-PMoA model average and combined PMoA models. Combined-PMoA models could differ in specific cases with respect to the concentrations at which they predicted a lower population-level sensitivity compared with the individual level (e.g., Cu toxicity to *D. magna*, Zn toxicity to *D. longispina*, independent of endpoint; Figure 8) and with respect to the

extent to which they predicted a lower population-level sensitivity compared with the individual level (e.g., Cu toxicity to *D. pulex*, Ni toxicity to *D. magna* and *D. longispina*, Zn toxicity to *D. pulex*; Figure 8). Because the population-level sensitivity in the present study is expressed as a difference to the individual-level sensitivity predicted by the same model, differences between predictions indicate different model behavior emerging on the population level.

DISCUSSION

PMoAs across species and metals

Our results suggest that the assumption of identical PMoAs may be defensible for Ni and Zn, but not for Cu. Reproduction efficiency was identified as the most likely PMoA for Ni and Zn, independent of the composition of the calibration data or outcome of prior statistical analysis. This was despite the fact that significant effects of Zn on 21 day carapace length were

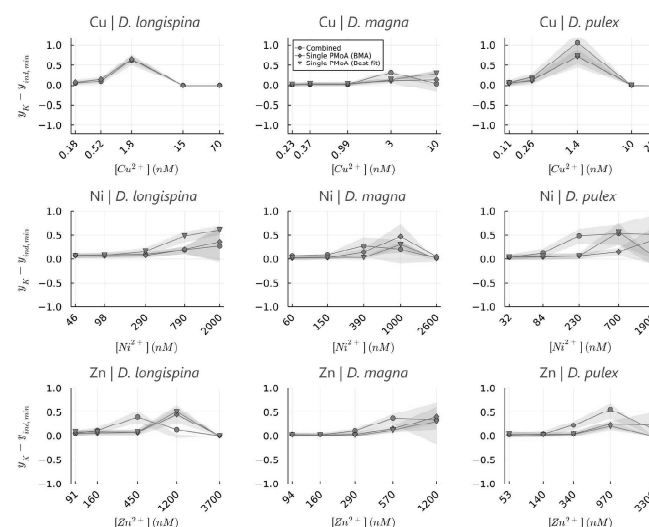


FIGURE 8: Predicted responses of carrying capacity (y_k), expressed as difference to the most sensitive individual-level response (final cumulative reproduction or survival) predicted by the same model ($y_{k,min}$). Positive values indicate lower sensitivity at the population level. BMA = Bayesian model average. PMoA = physiological mode of action.

observed in *D. longispina* and *D. pulex*, suggesting that an improved fit of PMoA reproduction efficiency to the reproduction data outweighed the prediction of effects on carapace length. It should be noted that the distance function used to calibrate the models was designed to assign the same weight to each measured variable and treatment. Therefore, the fact that fewer observations were available for growth compared with reproduction does, on its own, not explain this result, although data on effects on somatic growth rates is helpful to differentiate between different PMoAs that affect growth.

For Cu, the most probable PMoAs were assimilation efficiency and reproduction efficiency (*D. magna*) and embryonic investment (*D. pulex*). The biological interpretation of variability in PMoAs across species and metals requires great care because the link between PMoAs and traditionally considered toxic modes of actions, or even molecular initiating events, is not trivial. We speculate that interspecific variability in PMoAs for Cu might be the result of multiple molecular initiating

events leading to adverse effects (Brix et al., 2022). Interspecific variability in physiological traits, such as sodium turnover rates or resistance to oxidative stress (Barata et al., 2005), might shift the relative importance of molecular initiating events and lead to qualitatively different responses on the individual level, which in turn lead to interspecific variability in the estimated probabilities of PMoAs. If this is the case, PMoAs of Cu might not only be variable between species, but also within species or even individuals because the relevant physiological processes can also be influenced by environmental factors (Glover et al., 2005). This would also further support the choice of modeling approaches which do not make a hard selection for a single PMoA.

When only considering the most probable PMoAs for each species/metal combination, the PMoAs identified in the present study were different from those identified in other studies.

In *D. magna*, Vlaeminck et al. (2021) have not observed sublethal effects of Zn. The omission of a sublethal TKTD

submodel for Zn by Vlaeminck et al. (2021) occurred on the basis of a prior statistical analysis and possibly reflects inter-locational variability or differences due to different test media. Vlaeminck et al. (2021) also identified assimilation efficiency as PMoA of Cu in *D. magna*, which might also be a result of differences in the experimental test system, but could in addition be caused by different modeling assumptions and calibration approaches. However, the PMoAs assimilation efficiency for Cu and growth costs for Ni were also among the most probable in *D. magna* according to our analysis, and were not generally ruled out. This is also the case for Ni toxicity to *D. magna*, where Pereira, Vlaeminck, et al. (2019) identified growth costs as PMoAs.

The differences to results from other studies could therefore also result from epistemic uncertainty. The identification of reproduction efficiency as dominant PMoA for Cu, Ni, and Zn in *D. magna* is, however, in line with the outcome of a population-level analysis that used a combined-PMoA approach (Hansul et al., 2021).

For the *D. magna* dataset, it can in addition be said that the lack of length measurements in the metal treatments likely introduced additional uncertainty in the selection of PMoAs. Also, length and reproduction in the control was lower than usually observed in *D. magna*, and would not have fulfilled standard testing criteria according to OECD 2012). Some of the variability between studies may be caused by variable performance in the controls. Beyond that, we do not consider the lower baseline growth and reproduction of *D. magna* in the presented dataset to be a problem in general because the aim of the present study was not to provide standard toxicity test results and the used lineages were not selected to fulfil OECD criteria. Furthermore, the natural intraspecific variability in growth and reproduction is of interest for research purposes that go beyond standard toxicity testing, especially in the context of population- and community-level effects.

Comparison of modeling approaches

We found that single-PMoA models fit individual-level data overall equally well compared with combined PMoA models, despite the fact that the combination of multiple PMoAs in principle offers higher model flexibility. This likely reflects the fact that combined PMoA models are also more prone to suffer from the curse of dimensionality, that is, a steep increase in the computational costs for calibration with increasing number of parameters (Sunnåker et al., 2013). Because computational resources are finite, this may explain a less-than-expected increase in the performance advantage of combined PMoA models. Furthermore, there lies an additional challenge in defining meaningful priors for such models. We found in four of eight cases (Cu toxicity to *D. magna*, Zn toxicity to all species) that combined PMoAs performed better at predicting effects on reproduction at low effect levels, but this was not reflected in the global assessment of goodness-of-fit. A noteworthy difference between combined PMoA and single-PMoA models was the predicted effect on unmeasured variables (crasped length in

D. magna), where combined-PMA models implied stronger effects and narrower prediction intervals. Because combined-PMA models require a larger number of parameters to be fitted simultaneously, but explained the data overall equally well, the single-PMA models are more appropriate from a pure model calibration perspective. Limitations of single-PMA models can to some extent be dealt with using Bayesian model averaging. This is particularly useful in cases where the relative likelihoods of the most likely PMAs are close to each other because it is not necessary to make a hard selection for an individual PMA to apply the model. Bayesian model averaging can furthermore aid in dealing with limitations in the data itself, for example, if not the full concentration-response curve was observed.

Regarding the global goodness of fit, we found combined-PMoA models preferable for Ni and Zn models for *D. pulex* due to specific time-dependent effects of reproduction, which were only reproduced with a combination of PMoAs. These effects only became visible by considering the entire time series and would not be captured by a standard statistical analysis of reproduction data because the final cumulative reproduction was not clearly affected at the same concentrations. The fact that this difference in performance between single-PMoA and combined-PMoA models was not reflected in quantitative performance metrics also highlights the importance of combining multiple evaluation tools (i.e., multiple quantitative criteria and visualizations) to judge the performance of models.

To formalize the process of defining priors for TKTD model calibration, the empirical Bayes approach, that is, the utilization of a simplistic statistical analysis to define weakly informative priors, has proven useful. This process can be used to make sure that the same rationale is applied to different species and substances. Empirical Bayes further helps to avoid uniform prior distributions, which can be otherwise hard to avoid if no prior information is available. Due to the hard limits of uniform priors, defining uniform priors conservatively can be even harder than for normal or log-normal distributions. Furthermore, uniform priors are not always available for all parameters. For example, it is hard to justify that in any case, $\kappa = 0.3$ should have the same probability as $\kappa = 0.6$, but that the probability of $\kappa = 0.29$ should simultaneously be 0. Another motivation to use empirical Bayes stems from the fact that the prior probability of parameter values can be very tightly connected to the mathematical model (Gelman et al., 2017). Therefore, even if a large amount of data is available (e.g., on chronic Cu toxicity in *D. magna*), this data is only useful as prior information if it has been analyzed in the light of a model that follows the same basic principles. Consequently, the amount of useful prior information for DEB-TKTD models is usually much smaller than the amount of empirical data. A noteworthy potential drawback of empirical Bayes is that it can overstate the degree of uncertainty in the posterior distribution of the parameters (e.g., parameter values or PMoAs) because the data are used twice in the process of deriving probabilities (once to define priors, once to conduct the calibration). Therefore, if independent prior information is available in a form that is useful for the model in question, we consider their usage still to be

considered was quite small, however, and this was in turn motivated by the potential for strong correlations between the TK and TD parameters. That is, slow TK in combination with highly sensitive TD can lead to very similar effect patterns to fast TK and less sensitive TD. For the application of DEB-TKTD models to the population level, we suggest that an important next step to be taken is further model analysis in the form of global sensitivity analysis for TKTD parameters on the population level to assess to what extent possible non-identifiability of TK parameters affects predictions of effects on population-level effects. We hypothesize that this might in turn be dependent on the scenario (e.g., constant vs. time-variable exposure) as well as the endpoint (e.g., long-term effects on population density vs. time to recovery).

Population-level sensitivity

Combined-PMoA models predicted overall higher population-level sensitivity compared with the single-PMoA model average. This demonstrates that models with overall equal goodness-of-fit can still lead to substantially different conclusions. The predictions based on the best fit also differed substantially from the single-PMoA Bayesian model average (BMA) in specific cases, which illustrates that the propagation of uncertainty in mechanistic effect models is not only a matter of constructing prediction intervals. Instead, a modeling approach which includes uncertainties can also deliver a different overall prediction.

Populations were predicted to be equally or less sensitive than individuals. This prediction is in line with experimental evidence of the population-level sensitivity of *D. magna* to metals (Pereira, Blust, et al., 2019; Vlaeminck et al., 2021). Our modeling study therefore suggests that standard individual-level chronic endpoints (cumulative reproduction and survival) are conservative for single-metal population effects across *Daphnia* species under the scenario that was considered in our simulations. This general conclusion is independent of the tested modeling choices and population-level endpoint (growth rate or carrying capacity).

By expressing population-level sensitivity as a difference to the individual-level sensitivity predicted by the same model, we found that the differences between predictions were to a large extent caused by qualitatively different predictions of individual-level sensitivity. However, predictions differed with respect to the extent of the difference between population-level and individual-level sensitivity, and the concentrations at which the differences occurred. This signals different model behavior emerging at the population level at individual concentrations, which can be caused by different relative contribution of PMoAs to the predicted effect or by different relative contributions of lethal and sublethal toxicity to the predicted effect. Based on the difference between individual-level and population-level sensitivity, combined PMoA models were not generally more conservative, supporting the notion that this trend was chiefly caused by different predicted individual-level sensitivity, rather than mechanistic differences between the models.

CONCLUSIONS

Our results indicate that the assumption of an identical PMOA (reproduction efficiency) across *Daphnia* species is appropriate for Ni and Zn, but PMOAs of Cu likely vary within the genus. Based on global assessment of the goodness-of-fit, the use of single PMOA models is in general justified. However, we suggest that the propagation of uncertainty of both PMOA selection and parameter estimation should be routinely considered for DEB-TKTD models, given that uncertainty propagation can affect the prediction itself, not only provide prediction intervals. Combined PMOA models tended to predict higher population-level sensitivities, which was caused largely by a more conservative fit to the individual-level data. They may be considered as an option if effects at low concentrations are not predicted well by single PMOA models, but are likely only viable within a probabilistic modeling approach unless additional (physiological) data is available to further differentiate between combinations of mechanisms.

According to model predictions, standard chronic endpoints (cumulative reproduction or survival) are conservative for single-metal population effects across metals and species, independent of the tested modeling choices.

Supporting Information—The Supporting Information is available on the Wiley Online Library at <https://doi.org/10.1002/etc.5782>.

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 This article has earned both an Open Data and an Open Materials badge for making publicly available the digitally shareable data necessary to reproduce the reported results. The data is available at <https://doi.org/10.6084/m9.figshare.22930694>. Learn more about the Open Practices badges from the Center for Open Science: <https://osf.io/tvxvz/wiki/>.

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favorable over empirical Bayes, but an empirical Bayes approach in turn to be favorable over the use of uniform priors.

Implications of the PMoA "embryonic investment"

We used the PMoA embryonic investment to model metal toxicity to *D. pulex*, and it was identified as the most probable PMoA for Cu toxicity to *D. pulex* in the single PMoA models. This PMoA has to our knowledge not been used before to model toxic effects in any species or test substance within a DEB-TCX approach. It has interesting properties because it can explain the observed effects of Cu on *D. pulex* without a decrease in growth without implying a biologically implausible fitness increase. Any explanation of hormetic effects has to adhere to basic principles of mass and energy balance. Based on the categorization of hormetic effects used by Jager et al. (2013), an effect on embryonic investment can be categorized as allopathory hormesis, because it only influences the way energy is distributed, but does not imply that more energy is available (acquisitional hormesis). Acquisitional and mediational hormesis could be ruled out: any mechanism that could cause acquisitional hormesis, namely an increase in assimilation efficiency or ingestion rate, would also result in corresponding increases in body size, which has not been observed. Mediational hormesis was ruled out because at the same concentrations at which hormetic effects were observed, clear negative effects on survival were observed, and an increase in reproduction likely exceeded essential requirements in *Daphnia* by a fair margin (Kilham et al., 1999).

We present the PMoA embryonic investment as a testable hypothesis to explain some effects that are not predicted by the four typically considered PMoAs. The relevance of this PMoA can be empirically tested more directly, for example by measuring egg dry mass or energetic content. If embryonic investment is a relevant mechanism, the dry mass of eggs or energetic content should be decreased to an extent that is not predicted via indirect effects that can result from other PMoAs, in combination with the corresponding predicted (absence of) effects on brood sizes and carapace length. Data on the size of the brood and the energetic content of the eggs can be used to test this PMoA, but we are not aware of any study that has done this. It should be noted that the stress coefficient associated with this PMoA cannot vary freely because an embryo cannot be initialized with an infinitely small amount of reserves. The inclusion of a maximum stress parameter is therefore necessary.

Limitations of current approaches

While DEB-TKTD models have already proven useful for the prediction and explanation of population-level effects, some limitations persist within the most commonly used approaches.

According to our results, reduction in assimilation efficiency was identified as a relevant PMoA in three cases (Cu toxicity to *D. magna* and Zn toxicity to *D. longispina* and *D. pulex*). A current limitation with respect to this PMoA is that with basic

life-table data (reproduction, growth, survival), effects on ingestion rates cannot be differentiated from effects on assimilation efficiency because they have identical downstream effects on energy budgets. On the population level, however, effects on ingestion rate and assimilation efficiency can imply different dynamics due to the feedback with environmental food concentrations. It could therefore also be expected that effects on ingestion rate and assimilation efficiency will lead to different population-level effects of metals in combination with other environmental factors, such as food availability, and will differ in the resulting effects on multitrophic systems. Direct measurements of ingestion rate are time-intensive and not always trivial to realize, and therefore, it is realistically better to include routine life-table experiments. An efficient strategy may be to first develop the model based on regular life-table data and take additional measurements within a reduced experimental setup if reduction in feeding rate or assimilation efficiency are identified as relevant mechanisms. This may be based on easier-to-measure proxies such as limb beat rate (Lampert & Brendelberger, 1996). Reduced ingestion rates have been experimentally observed as a function of Ni exposure in *D. magna* (Ferreira et al., 2010), but because any mechanism that adversely affects body size will also indirectly affect ingestion rates, it cannot be directly concluded from this observation that a mechanistic effect on ingestion rates occurred. For this, mechanistic modeling, or at least a correction for body size in the statistical analysis, is necessary. The decision to use different models for the different endpoints (ingestion efficiency and ingestion rates would further lend itself to application of Bayesian methods because the posterior probabilities of the first development step can be used to inform the refinement of the model in a second step.

Selection of PMoAs with single PMoA models was associated with varying degrees of uncertainty. We see this as a reason to revisit the common practice of making a hard selection of a single PMoA based on the lowest goodness-of-fit only.

We suggest that the Bayesian model average should be routinely considered as an option unless the identification is unequivocal. In our analysis a hard selection of a single PMoA would have been justified in only two out of nine cases (Cu toxicity to *D. longispina*, Ni toxicity to *D. pulex*) when using a strict threshold (e.g., 5% probability of all other PMoAs), and possibly in one additional case (Zn toxicity to *D. magna*) when using a less strict threshold (e.g., most probable PMoA at least twice as probable as the second-most probable PMoA).

For combined-PMoA models, we conclude that it is probably not possible to identify a combination of PMoAs as the most probable model because alone the identification of single PMoAs was associated with considerable uncertainty. Under these circumstances, we suggest that the application of combined-PMoA models should be combined with a probabilistic modeling approach, where all uncertainties are propagated.

Another limitation occurred in the estimation of the dominant rate constant $k_{D,z,j}$ from effect data because we found that the difference between prior and marginal posterior distributions was overall very small. The range of values

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